

Linear algebra and geometry 1 - TTPU

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State Finished

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Time taken 34 secs

Marks -150.00/1000.00

Grade -4.95 out of 33.00 (-15%)

Question 1

Incorrect

Mark -15.00 out of
100.00

Fixed a coordinate system in the space, consider the points of coordinates $P_0 = (0, 0, 0)$, $P_1 = (2, 0, 1)$ e $P_2 = (1, 1, 1)$. Then, the area of the triangle of vertices P_0 , P_1 and P_2 is equal to

- ☐ (a) $\frac{\sqrt{6}}{2}$.
- ☒ (b) $\sqrt{6}$. ✖
- ☐ (c) $2\sqrt{3}$.
- ☐ (d) 6.

Risposta errata.

The correct answer is: $\frac{\sqrt{6}}{2}$.

Question 2

Incorrect

Mark -15.00 out of
100.00

Consider the following line $r : x - y = x + y - z = 0$.

Find the correct statement.

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- ☐ (a) r lies on the plane $z = 0$.
- ☒ (b) r passes through the point $(1, 1, 0)$. ✖
- ☐ (c) r lies on the plane $2x + z = 0$.
- ☐ (d) r has the same direction of the vector $(1, 1, 2)$.

Risposta errata.

The correct answer is: r has the same direction of the vector $(1, 1, 2)$.

Question 3

Incorrect

Mark -15.00 out of
100.00

Consider the following linear system in the variables x, y, z :

$$\begin{cases} x - y + z = 3 \\ x + z = 4 \\ 2x - 3y + 2z = 5 \end{cases}$$

Find the right statement.

-
- ☐ (a) The system has one unique solution.
- ☒ (b) The system has only two solutions. ✖
- ☐ (c) The system has infinitely many solutions.
- ☐ (d) The system has no solutions.

Risposta errata.

The correct answer is: The system has infinitely many solutions.

Question 4

Incorrect

Mark -15.00 out of
100.00

Let us take the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix}$; find the right statement:

- ☐ (a) $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$.
- ☐ (b) $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$.
- ☒ (c) $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix}$. ✖
- ☐ (d) $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix}$.

Risposta errata.

The correct answer is: $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$.

Question 5

Incorrect

Mark -15.00 out of
100.00

Consider a linear mapping $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$. Find which one of the following statements cannot hold.

- ☐ (a) f is injective.
- ☐ (b) The matrix associated to f has rank 1.
- ☒ (c) $\dim \operatorname{Im}(f) = 2$. ✖
- ☐ (d) $\dim \ker(f) \geq 1$.

Risposta errata.

The correct answer is: f is injective.

Question 6

Incorrect

Mark -15.00 out of 100.00

Let us consider the vectors $\vec{v}_1 = (1, 1, 1, 0)$, $\vec{v}_2 = (2, 2, 0, 2)$, $\vec{v}_3 = (3, 0, 3, 3)$ e $\vec{v}_4 = (0, 4, 4, 4)$ in $\mathbb{R}^4$. Find the right statement.

- ☐ (a) $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ is a basis of $\mathbb{R}^4$.
- ☐ (b) The vector $\vec{v}_1$ lies in the subspace generated by $\{\vec{v}_2, \vec{v}_3, \vec{v}_4\}$.
- ☐ (c) $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis of $\mathbb{R}^2$.
- ☒ (d) There exist $a, b, c \in \mathbb{R}$ such that $\vec{v}_4 = a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3$. ✖

Risposta errata.

The correct answer is: $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ is a basis of $\mathbb{R}^4$.

Question 7

Incorrect

Mark -15.00 out of 100.00

Let $q(x, y) = x^2 - 8xy - y^2$ be a quadratic form. Find the correct statement.

- ☐ (a) The matrix associated to q is $\begin{pmatrix} -1 & -4 \\ -4 & 1 \end{pmatrix}$.
- ☒ (b) q is negative definite. ✖
- ☐ (c) q is positive definite.
- ☐ (d) q is indefinite.

Risposta errata.

The correct answer is: q is indefinite.

Question 8

Incorrect

Mark -15.00 out of 100.00

Let us consider the conic $\mathcal{C}$ in the Euclidean plane defined by the equation $2x^2 + 2xy + 2x + 2y^2 - 1 = 0$.

Find the correct statement.

- ☐ (a) $\mathcal{C}$ is a parabola.
- ☒ (b) $\mathcal{C}$ is degenerate. ✖
- ☐ (c) $\mathcal{C}$ is a hyperbola.
- ☐ (d) None of the three other statements is correct.

Risposta errata.

The correct answer is: None of the three other statements is correct.

Question 9

Incorrect

Mark -15.00 out of
100.00Let $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ such that $f(x, y, z) = (2x + 2y + 2z, 2x + 2y + 2z, 2x + 2y + 2z)$. Let λ_i be an eigenvalue of f and let $E(\lambda_i)$ be the corresponding eigenspace.

Find the correct statement.

- ☐ (a) $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 2, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (b) $\lambda_1 = 2, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☒ (c) $\lambda_1 = 6, \lambda_2 = 2, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$ ✖
- ☐ (d) $\lambda_1 = 6, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☐ (e) $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (f) $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2, \dim E(\lambda_3) = 1$

Risposta errata.

The correct answer is: $\lambda_1 = 6, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$

Question 10

Incorrect

Mark -15.00 out of
100.00

For $a, b \in \mathbb{R}$ consider the following linear system in the variables x, y, z :

$$\begin{cases} x - ay + z = 0 \\ x + z = 0 \\ bx - y + 2z = 0 \end{cases}$$

and let $S_{a,b}$ be the set of solutions of the linear system of equations.

Find the right statement.

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- ☐ (a) For $b = 3$ and $a = 1$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 2$.
 - ☐ (b) For $b = 3$ and $a = 2$ the linear system does not have solutions.
 - ☒ (c) For $b = 3$ and $a = 2$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 3$. ✖
 - ☐ (d) For $b = 1$ and $a = 0$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 2$.
 - ☐ (e) For $b = 3$ and $a = 1$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.
 - ☐ (f) For $b = 1$ and $a = 0$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.

Risposta errata.

The correct answer is: For $b = 1$ and $a = 0$ the linear system admits infinitely many solutions and $\dim S_{a,b} = 1$.